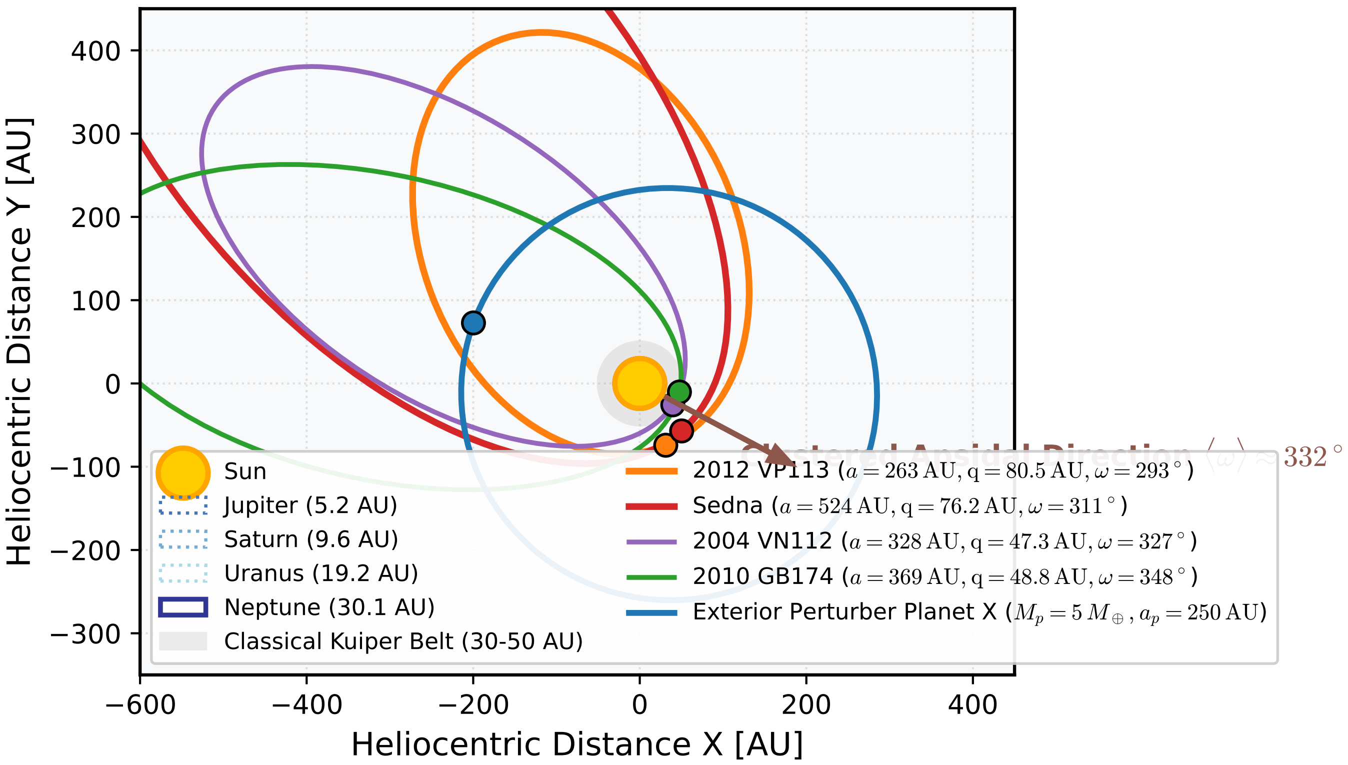
(a) Orbital Architecture of Extreme TNOs, Sednoids, and Exterior Perturber



PROPOSED ORIGIN MECHANISMS FOR DETACHED SEDNOIDS (q > 70 AU):

1. Secular Kozai-Lidov Resonance with Exterior Planet (T&S 2014):
   • Super-Earth (Mp ~ 2-15 M_Earth) at ap ~ 200-400 AU.
   • Induces stable libration of omega around 0° / 340° (tau_lib ~ 200-400 Myr).
   • Explains BOTH high perihelion (e-i exchange) AND persistent omega clustering

2. Early Solar Birth Cluster Stellar Encounter (Morbidelli & Levison 2004):
   • Passing star (M* ~ 0.8 M_sun, b ~ 400 AU) in embedded cluster (t < 50 Myr).
   • Impulsive tidal torque lifts q from 30 AU to > 70 AU.
   • Limitation: Leaves omega to randomize over 4.5 Gyr under giant planet preces

3. Rogue Planet Scattering (Gladman & Chan 2006):
   • Ejected proto-planet transiently lifts Sednoids before hyperbolic escape.
   • Limitation: Randomizes omega on Gyr timescales post-ejection.

MATHEMATICAL SECULAR FREQUENCY BALANCE:

Giant Planet Quadrupole Precession:
  d omega / dt |_giants = + (3/4) n [sum (M_k a_k^2) / (M_sun a^2)] * F(e, i)
  where F(e, i) = (5 cos^2 i + 2 cos i - 1) / [2 (1 - e^2)^2] > 0
  -> Drives rapid prograde precession (tau_prec ~ 30 - 1000 Myr).

Exterior Perturber Kozai Torque:
  d omega / dt |_pert = + (3/4) (Mp / M_sun) (a / ap)^3 [n / sqrt(1 - e^2)]
                      * [(5 cos^2 I_mut - 1) + 5 sin^2 I_mut cos(2 omega)]

Resonant Equilibrium & Libration Center:
  d omega / dt |_total = d omega / dt |_giants + d omega / dt |_pert = 0
  -> Creates stable libration islands centered at omega_0 ~ 0° / 340°!
  -> Confines eTNO perihelia near the ecliptic plane.